

PURVAV PUNYANI

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SUMMARY

Software Engineer with 2 years full-time on the SSA Anti-Fraud program, modernizing legacy SAS fraud models into Python data pipelines on AWS that move 10M+ records a day. Earlier data science at GreenPortfolio and client automation at HighRadius, plus AI products built at hackathons. Comfortable scoping work directly with non-technical stakeholders and owning it through production.

EXPERIENCE

Software Engineer, Leidos Inc. | Data Architect/Engineer, SSA Anti-Fraud

Sep 2024 - Present

SSA Anti-Fraud program: fraud-detection models that flag suspicious activity for SSA analysts and investigators.

- Ported legacy SAS fraud models to Python, building pre-fraud and sampling modules and retiring 5 services and 2 pipelines
- Built an AWS Athena batch pipeline (Parquet, bulk COPY) loading 21M telecom records/month in 30 min for fraud analytics
- After a legacy data source was retired, rebuilt a claims fraud model on payment data, cutting its inputs from 12 tables to 3
- Scoped new data needs in recurring SSA analyst meetings, then designed schemas, queries, and pipelines for 3 fraud models
- Kept 10M+ records/day flowing to fraud models via Airflow/Lambda batch jobs on AWS EC2 (Greenplum, SQL Server, Postgres)
- Built a service that checks each job right after its expected finish time, flagging late or failed runs across 20+ production models

Data Science Intern, GreenPortfolio

Oct 2023 - Apr 2024

NYC climate-fintech startup helping individual investors align their portfolios with their values.

- Cut cloud cost 10% and processing time 20% on daily securities ETL (Cloud Functions, BigQuery) by tuning execution and storage
- Improved forecast accuracy 15% with a deployed forecasting pipeline whose explainable outputs show each prediction's drivers
- Raised financial-news insight relevance 25% by prototyping embedding-based clustering that groups related stories into themes

Software Engineering Intern, HighRadius Corporation

Jul 2021 - Jun 2022

B2B fintech SaaS automating accounts receivable and order-to-cash workflows for Fortune 500 finance teams.

- Automated claim retrieval from 3-4 client portals a week (incl. Starbucks, Walgreens) with Java/Selenium agents, archiving to S3

Machine Learning Engineer Intern, Grroom

Nov 2019 - Feb 2020

AI fashion startup helping shoppers match outfits and buy the pieces they pick.

- Trained YOLO clothing detectors on NVIDIA GPUs from 200+ labeled images per category to power the app's outfit matching

PROJECTS

CineFIBO | Bria FIBO Hackathon | Next.js, TypeScript, FastAPI, OpenAI, Bria FIBO

Dec 2025 | [Demo](#) | [Code](#)

- Turned one scene into a coverage plan of up to 12 shots for creators by pairing OpenAI shot planning with Bria FIBO renders
- Replaced trial-and-error prompting with 4 structured controls (angle, lens, mood, color) that re-render a shot predictably

AI Voice Banking Assistant | TikTok TechJam 2024 | Flask-SocketIO, React, Whisper, GPT-4, MySQL

Jul 2024 | [Demo](#) | [Code](#)

- Wrote 44 of 75 commits (team of 2) on a voice assistant resolving 8 banking requests, from balance checks to fraud flags
- Built 4-step caller authentication and real-time Socket.IO task handling across 11 intents, saving call summaries to MySQL

Zippy: In-Memory Key-Value Store | team of 3 | C++, gRPC, Protobuf, GoogleTest

Jun 2024 | [Code](#)

- Built the async gRPC request path (completion queue, 3 worker threads); store hit ~21K ops/sec, 0.36 ms avg GET at 10 clients
- Added per-client request tracking, timestamped logs (terminal or file), and concurrency tests to a 13-test GoogleTest suite

Data Poisoning Detection | NYU paper, team of 2 | TensorFlow, VGG16, scikit-learn, Docker

Dec 2023 | [Paper](#) | [Code](#)

- Co-authored a study of 4 data-poisoning attacks on a VGG16 CIFAR-10 model; poisoned training cut accuracy to as low as 15%
- Built the VGG16 baseline, 2 attack pipelines, and Isolation Forest detection (73% accuracy vs 67% for One-Class SVM)

SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, SAS (migration)

Backend: FastAPI, Flask, gRPC, Socket.IO, REST API design, Selenium, Docker, Git

Data: ETL/ELT pipelines, data modeling, Airflow, Parquet, Greenplum, PostgreSQL, SQL Server, MySQL, BigQuery

AI / ML: OpenAI APIs, LangChain, RAG, embeddings, semantic routing, Whisper, YOLO, PyTorch, TensorFlow, scikit-learn

Cloud & Ops: AWS (EC2, S3, Athena, Lambda), GCP (BigQuery, Cloud Functions), APScheduler, monitoring and alerting

Frontend: React, Next.js, TypeScript, HTML/CSS

EDUCATION

New York University - M.S. Computer Engineering | Merit Scholarship | GPA 3.82/4.0

New York, NY | Sep 2022 - May 2024

Course Assistant, Computer Networking (2 semesters): supported 40+ students through lectures, twice-weekly office hours, and grading

KIIT University - B.Tech Computer Science and Engineering | GPA 8.98/10

Bhubaneswar, India | Jul 2018 - Jun 2022